

NLP Report — Explainable Sentiment Analysis using BERT

A Transformer-Based Sentiment Classification Framework using Attention Visualisation, SHAP, and LIME

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ABSTRACT

The rapid growth of online platforms and social media applications has significantly increased the volume of textual data generated daily. Sentiment analysis systems are therefore essential for understanding public opinions, emotions, and customer behaviour automatically. This research presents an explainable sentiment analysis framework using the BERT transformer architecture integrated with Attention Visualisation, SHAP, and LIME explainability methods. The Amazon Polarity dataset was used for binary sentiment classification, and the model was fine-tuned using Hugging Face Transformers and PyTorch. Experimental evaluation demonstrated strong classification performance with approximately 90% accuracy and F1-score. Additionally, explainability techniques provided transparent and interpretable insights into transformer prediction behaviour. The study contributes a reproducible NLP pipeline that balances predictive performance and model transparency.

1. INTRODUCTION

1.1 Background

Natural Language Processing (NLP) has become one of the most rapidly advancing fields within artificial intelligence. Applications such as sentiment analysis, question answering, machine translation, and text summarisation rely heavily on NLP techniques. Among these applications, sentiment analysis is widely used for customer feedback monitoring, recommendation systems, and social media analysis.

1.2 Problem Statement

Although transformer-based architectures achieve high accuracy, they are often criticised for their black-box behaviour. Users cannot easily understand why the model predicts a specific sentiment category. This lack of transparency reduces trustworthiness and interpretability.

1.3 Research Objectives

The objectives of this study include: • Implementing a transformer-based sentiment analysis framework using BERT. • Fine-tuning the pretrained model on Amazon Polarity dataset. • Integrating SHAP, LIME, and Attention Visualisation. •

Evaluating classification performance using standard metrics. • Analysing transformer interpretability and prediction behaviour.

1.4 Contributions of the Study

This research contributes a reproducible NLP pipeline capable of balancing predictive performance and explainability. The integration of multiple interpretability techniques provides detailed insights into transformer decision-making.

2. RELATED WORK

2.1 Classical Machine Learning Approaches

Early sentiment analysis systems relied on statistical machine learning algorithms such as Naive Bayes, Logistic Regression, and Support Vector Machines. These methods used handcrafted features including Bag-of-Words and TF-IDF vectors.

2.2 Deep Learning Approaches

Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Long Short-Term Memory (LSTM) architectures significantly improved sequential representation learning. However, these methods still struggled with long-range contextual dependencies.

2.3 Transformer-Based Models

The introduction of transformer architectures revolutionised NLP research. BERT introduced bidirectional contextual learning using self-attention mechanisms and achieved state-of-the-art results across multiple NLP tasks.

2.4 Explainable AI Research

Explainable AI methods such as SHAP and LIME have become increasingly important for interpreting complex deep learning models. Attention heatmaps additionally provide visual understanding of transformer focus distribution.

3. DATASET DESCRIPTION

3.1 Dataset Source

The Amazon Polarity dataset was loaded using the Hugging Face datasets library. The dataset contains product reviews labelled as positive or negative sentiment.

3.2 Dataset Configuration

The notebook used approximately: • 15,000 training samples • 3,000 testing samples • Binary sentiment labels • Random seed = 42

3.3 Data Preprocessing

Several preprocessing operations were applied: (i) Tokenisation using BERT tokenizer (ii) Sequence padding and truncation (iii) Tensor conversion using PyTorch (iv) Dataset shuffling and splitting

3.4 Tokenisation Process

The bert-base-uncased tokenizer converted input text into transformer-compatible token IDs while preserving contextual relationships between words.

4. METHODOLOGY

4.1 Model Selection

The bert-base-uncased transformer model was selected because of its strong contextual representation learning capabilities and state-of-the-art performance across NLP benchmarks.

4.2 Fine-Tuning Process

The pretrained BERT model was fine-tuned using Hugging Face Trainer API. Fine-tuning adapts pretrained contextual knowledge to sentiment classification tasks.

4.3 Attention Visualisation

Attention heatmaps were generated using transformer attention outputs. These heatmaps highlighted influential words contributing to prediction decisions.

4.4 SHAP Implementation

SHAP explanations quantified token-level contribution scores using Shapley values. Positive values increased sentiment probability while negative values reduced it.

4.5 LIME Implementation

LIME generated local surrogate explanations for individual predictions. The LimeTextExplainer module identified influential tokens affecting sentiment classification.

5. TRAINING PROCEDURE

5.1 Training Configuration

TrainingArguments included epoch-wise evaluation, checkpoint saving, batch size configuration, and logging settings.

5.2 Hyperparameters

• Model: bert-base-uncased • Batch Size: 4 • Epochs: 2 • Sequence Length: 64 • Framework: PyTorch + Transformers • Hardware: CUDA GPU

5.3 Optimisation

The AdamW optimiser was used for gradient optimisation due to its effectiveness for transformer architectures.

5.4 Model Convergence

Training loss consistently decreased across epochs, indicating successful convergence and optimisation.

6. EXPERIMENTAL RESULTS

6.1 Evaluation Metrics

The model was evaluated using accuracy, precision, recall, and F1-score metrics. These metrics were computed using scikit-learn functions implemented inside the notebook.

Metric	Score
Accuracy	89.97%
Precision	89.67%
Recall	90.67%
F1-Score	90.17%

6.2 Misclassified Sample Analysis

The notebook additionally analysed misclassified samples by comparing predicted labels with true labels. This analysis helped identify cases involving contextual ambiguity and complex sentiment expressions.

7. EXPLAINABILITY ANALYSIS

7.1 Attention Heatmaps

Attention heatmaps demonstrated how transformer layers distributed contextual focus across input tokens during sentiment prediction.

7.2 SHAP Explanations

SHAP explanations provided detailed token-level contribution analysis and enabled global interpretability.

7.3 LIME Explanations

LIME generated simplified local explanations for individual predictions and identified influential sentiment words.

7.4 Importance of Explainability

Explainability improves transparency, fairness, and user trust in transformer-based NLP systems.

8. DISCUSSION

The experimental findings demonstrate that transformer architectures significantly outperform traditional machine learning approaches for sentiment analysis tasks. BERT effectively captured contextual relationships through self-attention mechanisms. Explainability methods further improved model transparency without reducing predictive performance.

9. CONCLUSION

This study proposed an explainable sentiment analysis framework using BERT integrated with Attention Visualisation, SHAP, and LIME. Experimental evaluation demonstrated strong classification performance while maintaining

interpretability and transparency. The project contributes a reproducible NLP framework combining transformer performance with explainable AI methods.

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